



# Advanced Spa & Wellness Booking System

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## Project

Luxury-focused web booking platform for spas and wellness centres.

## Course

Final-year project — Computer Science / Web Development

## Design

Modern UI: soft pink, lavender, cream, white, light purple accents

# Introduction

The global spa and wellness industry is shifting online. Customers expect frictionless digital experiences for browsing services, checking therapist availability, and booking appointments. Web-based booking systems enable digital transformation, improve operational efficiency, and enhance the customer journey from discovery to confirmation.

- Industry overview and market readiness for digital services
- Importance of real-time booking and service discovery
- Benefits: convenience, reduced no-shows, centralised data







## Problem Statement

- Manual appointment scheduling is time-consuming and error-prone
- Customers lack easy visibility into services and therapist availability
- Operational inefficiency in managing overlapping bookings
- No centralised booking or customer management system

⚠ Current processes increase staff workload and reduce customer satisfaction.



## Project Objectives

### Online Booking

Build a secure, responsive booking interface accessible across devices.

### Service Catalogue

Present detailed service descriptions, durations and pricing.

### Therapist Profiles

Display skills, availability and ratings to help customer choice.

### Efficient Scheduling

Simplify appointments, confirmations and reduce manual effort.



# Technologies Used

Frontend stack chosen for speed, modularity and developer ergonomics.



## React JS

Component-driven UI for reusable views and stateful interfaces.



## JavaScript

Core language powering dynamic behaviour and client-side logic.



## CSS

Modern styling, responsive layouts, and theme colour application.



## React Router DOM

Client-side routing for multi-page single-page application flows.



## Local Storage

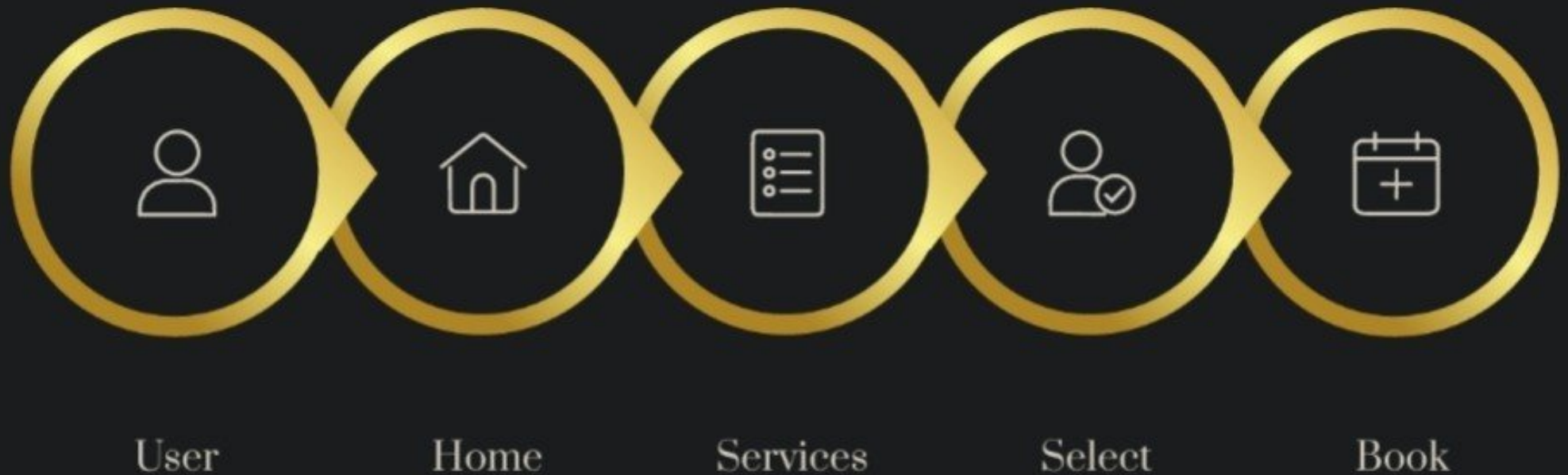
Persist bookings and user state locally during prototyping.



## Vite

Fast development server and build tooling for modern frontend apps.

# System Architecture



Architecture summary: single-page React app with routed pages, componentised UI, local persistence for bookings, and clear linear user journey from discovery to confirmation.



# Frontend Engineering Concepts Implemented

01

## Components

Modular UI pieces encapsulating markup and behaviour for reuse.

02

## Functional Components & JSX

Declarative UI using JSX syntax and pure function components.

03

## Props & State

Data flows via props; local state managed with useState for interactivity.

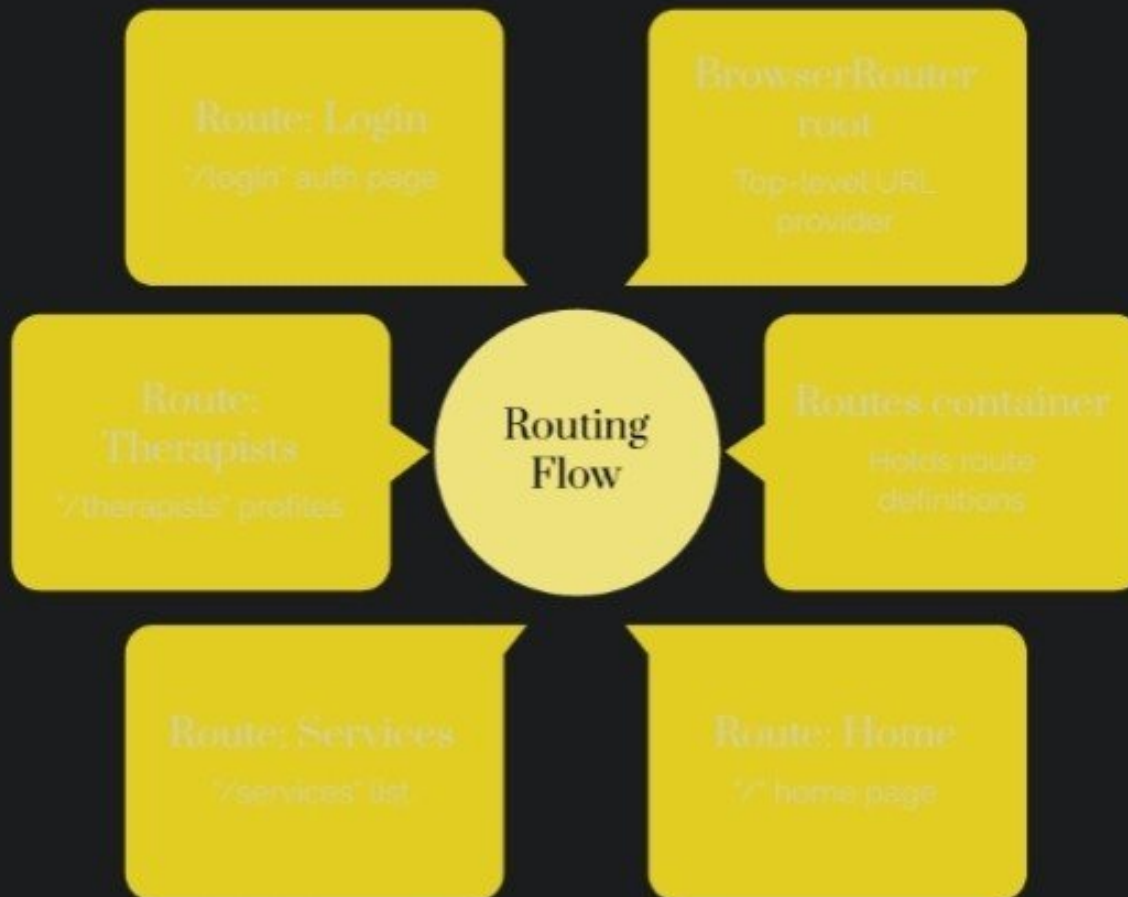
04

## Event Handling & Forms

Controlled components implement form inputs, validation and submission.



# Routing & Navigation



- **BrowserRouter** — top-level router providing URL-driven views
- **Routes & Route** — declarative route definitions for each page
- **useNavigate** — programmatic navigation after form submit or booking
- Pages: Home, Services, Therapists, Login, Booking — modular routes for clear UX

❶ Proper routing ensures deep links, back/forward behaviour and shareable URLs.



# Key Features



## Browse Services

Filterable service catalogue with durations, prices and descriptions.



## Therapist Profiles

Profiles include specialties, available slots and short bios.



## Online Booking

Step-by-step booking flow with confirmation and success popups.



## Responsive Design

Adaptive UI for mobile and desktop, maintaining luxury branding.

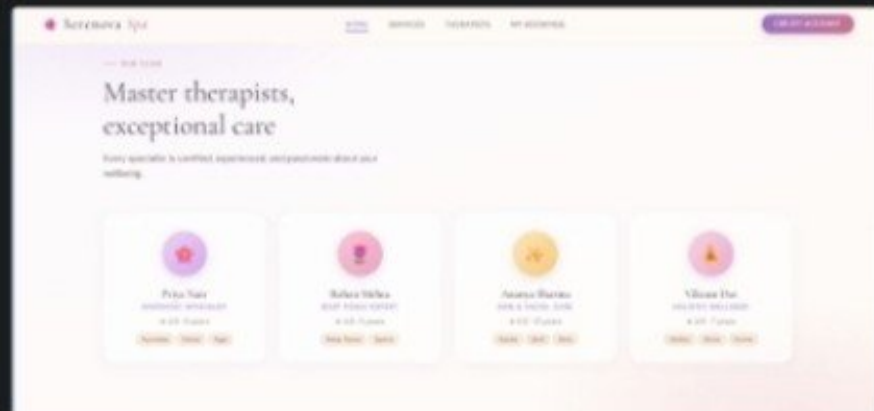
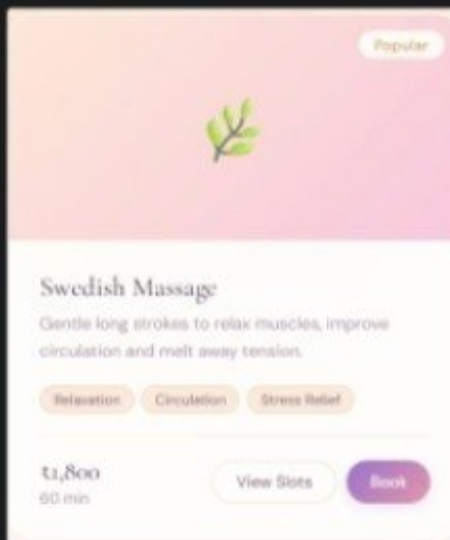
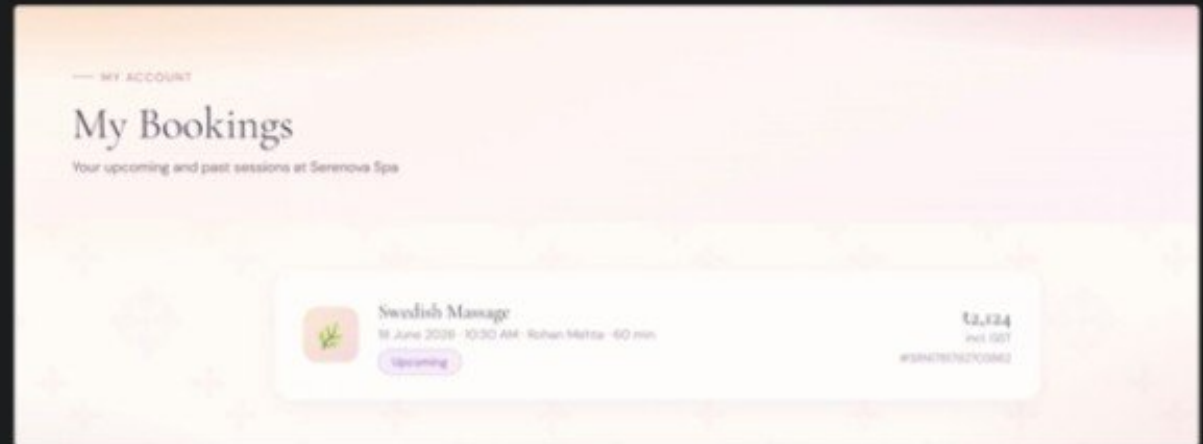
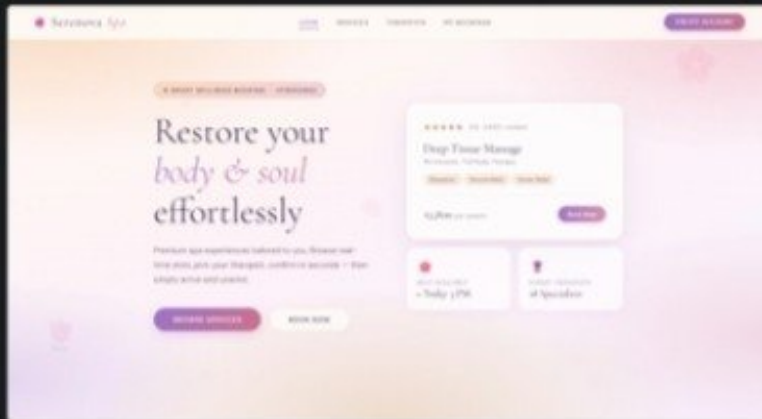


## Local Storage

Client-side persistence for bookings during prototype stage.

Other features: Login authentication flow, booking confirmation, service filters, success popups and lightweight client-side data management.

# User Interface Screens (Mockups)



The mockups follow the project's luxury aesthetic: soft pink, lavender, cream, white and light purple. UI prioritises clarity, accessible typography and micro-interactions for a premium feel.

# Advantages & Future Enhancements

## Advantages

Easy appointment booking; improved customer experience; reduced manual work; faster service management; responsive interface across devices.

## Planned enhancements

- Online payment gateway integration
- Email / SMS notifications for confirmations and reminders
- Admin dashboard with analytics and booking management
- Database-backed persistence (SQL/NoSQL)
- Real-time slot availability and conflict resolution
- Native mobile application for iOS/Android





## Conclusion

The Advanced Spa & Wellness Booking System demonstrates a polished, React-based front end that streamlines booking workflows and improves client access. The prototype validates core FED concepts while offering a clear roadmap for production features (payments, notifications, server-side storage).

- ❏ Project outcomes: functional booking flow, therapist profiles, local persistence, responsive UI — ready for extension to a full-stack implementation.